

Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."

2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - **The sending server's hostname/IP address.**
 - **The receiving server's hostname/IP address.**
 - **The date and time the email passed through.**

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.

- **Pass:** Indicates the IP is authorized.
- **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.

Example of a Header Analysis

Here's a quick example with a sample header:

vbnet

Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1])

by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

Received: by mail.example.com with SMTP id a1mr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.

cc Bina xcc Rear xcc Upp xcc Lear xgit h xGitH xmail xOrbi xOrig xSign xmx: xEma x

mxtoolbox.com/Public/Tools/EmailHeaders.aspx?huid=da38c120-4f86-43e7-b5ed-7cd0b8fde629

μTorrent Web

MX

TOOLBOX

SUPERTOOL

PricingToolsDelivery CenterMonitoringPr

SuperToolMX LookupBlacklistsDMARCDiagnosticsEmail HealthDNS LookupAnalyze Headers

Header Analyzed

Email Subject: Orbis Financial Corporation Ltd. Unlisted Shares

Delivery Information

DMARC Compliant

SPF Alignment

SPF Authenticated

DKIM Alignment

DKIM Authenticated

Relay Information

Received Delay:

0 seconds

From 180-244-smtp-out-us.ncmail01.com to mx.google.com

to

Your IP is: 103.238.730.130 | Contact Terms & Conditions Security API Privacy Phone: (866)-698-6652 | © Copyright 2004-2026, MXToolbox, Inc. All rights reserved. US Patents 10839353 B2 & 11461738 B2

SPF and DKIM Information

dmARC:promo.incredunlisted.com

Hide

Solve Email Delivery Problems

DMARC Record for promo.incredunlisted.com

No DMARC Record found for sub-domain.

Organization Domain of this sub-domain is: incredunlisted.com Inbox Receivers will apply incredunlisted.com DMARC record to mail sent from promo.incredunlisted.com

SP Tag '' found: Inbox Receivers will treat all mail sent from promo.incredunlisted.com that fails DMARC as suspicious.

DMARC Record for incredunlisted.com (organizational domain)

v=DMARC1; p=quarantine; rua=mailto:dmarcreport@incredunlisted.com; ruf=mailto:dmarcreport@incredunlisted.com; fo=1;

Tag	TagValue	Name	Description
v	DMARC1	Version	Identifies the record retrieved as a DMARC record. It must be the first tag in the list.
p	quarantine	Policy	Policy to apply to email that fails the DMARC test. Valid values can be 'none', 'quarantine', or 'reject'.
rua	mailto:dmarcreport@incredunlisted.com	Receivers	Addresses to which aggregate feedback is to be sent. Comma separated plain-text list of DMARC URIs.
ruf	mailto:dmarcreport@incredunlisted.com	Forensic Receivers	Addresses to which message-specific failure information is to be reported. Comma separated plain-text list of DMARC URIs.
fo	1	Forensic Reporting	Provides requested options for generation of failure reports. Valid values are any combination of characters '01ds' seperated by '!'.

Test	Result
DMARC Record Published	DMARC Record found
DMARC Syntax Check	The record is valid
DMARC Multiple Records	Multiple DMARC records corrected to a single record.
DMARC Policy Not Enabled	DMARC Quarantine/Reject policy enabled
DMARC External Validation	All external domains in your DMARC record are giving permission to send them DMARC reports.

Dkim Public Record:

v=DKIM1; k=rsa; p=MIIBIjANBgkqhkiG9w0BAQFAAQCAQEAaBrzrFOTgo5SYTgnDTFKdVYH6oTQcAEJgq7fwkpgDsTo9y047dEPWZiQciUL9Cft8a0V0Jg2gmp16u+Qo0ccsbaKh36RkpY6h81ch0w+U1V2zW3chU/P/Kgy560h8eQ1268JhVTrHAOG543vmbXG6mYA3zkaCce2/84Z8JhXKE3pfJ6cdXETqcC37rds

Dkim Signature:

v=1; a=rsa-sha256; c=relaxed/relaxed; d=promo.incredunlisted.com; s=n2048; h=feedback-id:list-unsubscribe-post:list-unsubscribe:content-type:mime-version: subject:message-id:to:from:from:to:subject; bh=Jqh7FzrXoQ4sQeDWZd10xxokrp4oIUJ2YP257Pj8=; b=soJc20oPjYrB

Tag	TagValue	Name	Description
v	DKIM1	Version	Identifies the record retrieved as a DKIM record. It must be the first tag in the record.
k	rsa (Length: 2048 bits)	Key Type	The type of the key used by tag (p).
p	MIIBIjANBgkqhkiG9w0BAQFAAQCAQEAaBrzrFOTgo5SYTgnDTFKdVYH6oTQcAEJgq7fwkpgDsTo9y047dEPWZiQciUL9Cft8a0V0Jg2gmp16u+Qo0ccsbaKh36RkpY6h81ch0w+U1V2zW3chU/P/Kgy560h8eQ1268JhVTrHAOG543vmbXG6mYA3zkaCce2/84Z8JhXKE3pfJ6cdXETqcC37rdsF+qdo22et1Ewq+5dAAkHtV608q+0RnA2HyVp0W0D6AhEn7H2MuQz0hF8Tz1c+Af1w00XN1ckFSu0ANpuj0Eptaaq2FdlhjR8uBp16WmDxja0QL0eBnTF0R7vYH48AuC4hEMbzrQIDAQAB	Public Key	The syntax and semantics of this tag value before being encoded in base64 are defined by the (k) tag.

Test	Result
DKIM Record Published	DKIM Record found
DKIM Syntax Check	The record is valid
DKIM Public Key Check	Public key is present
DKIM Signature Syntax Check	The signature is valid
DKIM Signature Identifier Match	Signature domain match
DKIM Signature Alignment	Signature domain in alignment.
DKIM Signature Duplicate Tags	Signature tags are unique
DKIM Signature Expiration	The signature is not expired

dkim:env.etransmail.com:fnc

Hide

Dkim Public Record:

v=DKIM1; q=+; k=rsa; p=NI6H9AGC5g51b3QDEBAQUAA6NADCB1QK8gQ081+QZ8QE2zK3CXc8JQ489yRvqDqp++5x5yr8CdcAoeXULmG1zV7nLGcpUNKX0L5nzUoe8HtpeH7t1V0/Xcbat8uW3t3ueWGePE++KchZ5eYYGtUXZc7Z88hU/PstJN3KFRZQnxEWK3PbQu9x8gH7m6ZQ4De0d4+354a6wIDAQAB

Dkim Signature:

v=1; a=rsa-sha256; c=relaxed/relaxed; d=env.etransmail.com; s=fnc; h=feedback-id:list-unsubscribe-post:list-unsubscribe:content-type:mime-version: subject:message-id:to:from:from:to:subject; bh=Jqh7FzrXoQ4sQeDWZd10xxokrp4oIUJ2YP257Pj8=; b=AYzug+WNWcHSteaKs/Z

Tag	TagValue	Name	Description
v	DKIM1	Version	Identifies the record retrieved as a DKIM record. It must be the first tag in the record.
g	*	Granularity of the key	
k	rsa (Length: 1024 bits)	Key Type	The type of the key used by tag (p).
p	MI0EMH0GCG9g5t1b3QDEBAQUAA6NADCB1QK8gQ081+QZ8QE2zK3CXc8JQ489yRvqDqp++5x5yr8CdcAoeXULmG1zV7nLGcpUNKX0L5nzUoe8HtpeH7t1V0/Xcbat8uW3t3ueWGePE++KchZ5eYYGtUXZc7Z88hU/PstJN3KFRZQnxEWK3PbQu9x8gH7m6ZQ4De0d4+354a6wIDAQAB	Public Key	The syntax and semantics of this tag value before being encoded in base64 are defined by the (k) tag.
d	env.etransmail.com	SDID value	The SDID claiming responsibility for an introduction of a message into the mail stream.
	promo.incredunlisted.com	From Domain	The domain used in the From header field.

Test	Result	
DKIM Signature Alignment	Signature domain not aligned.	More Info
DKIM Record Published	DKIM Record found	
DKIM Syntax Check	The record is valid	
DKIM Public Key Check	Public key is present	
DKIM Signature Syntax Check	The signature is valid	
DKIM Signature Identifier Match	Signature domain match	
DKIM Signature Duplicate Tags	Signature tags are unique	
DKIM Signature Expiration	The signature is not expired	

Header Name	Header Value
Delivered-To	saikiranmutyala18@gmail.com
X-Received	by 2002.a05.620a.222c.b0.937.2ec9.c98 with SMTP id af79cd13be357-9373942667bmr409879985a.5.1787319423474; Fri, 21 Aug 2026 06:37:03 -0700 (PDT)
ARC-Seal	i=1; a=rsa-sha256; t=1787319423; c=none; d=google.com; s=arc-20260327; b=JbY32iY5X1E7LpPjUWwNexC8f6cTUjKcQ8hTLUwVlIdg3IAAmzhXv6ABdQevv DQFFvRTWCPh5hNliqM8SIRETTmZtVdfcN5CP3ACXAV73U2J7hpCUOGaxMmQeMV+zt0 ORJPkYl/TIkbcqpp9Z9ACYF8I77SXEHu49Nh6QXxvy4WqIEETXIBhKBk85KAro2SI GwYJnFRJQOPoUt8w+C+pqG6V9QOecfGpYnLwdjNWj2N9g+ks0J4FZBNhYJCDU ZBGraF2+HicdvHf753FSe9SIVFS/Tddv55+bye7nXfCfDgICGeRNLevhOCORLhyI UC/Q=
ARC-Message-Signature	i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=feedback-id:list-unsubscribe-post:list-unsubscribe:mime-version: subject:message-id:to:from:date:dkim-signature:dkim-signature; bh=Jqh7FzrXoQ4sQeDWZd10xxokrp4oIUJ2YP257Pj8=; fh=HcW2MgtVgQuC+HGSVYCTha9DXJKPKH0S2B8IYrRio=; b=OH20GaL+gYsQarp5aIlUdHFxQ2c9ulmXytlvOQc86UJ+dpks8DUJQKX0bpKcRIH1 L+9b3TzCfOMuNvcuN4K7P4dAlfapX3/74LnDyG4h29TKhshqOvW1ukM4euOotum CWfV/BGK4MhJB8YdarN2m0cNdaBn5rqQbtr0FJBC3S3KZKJuoEhCvoTbwMdxOFkJS+SCpkriNefJIDcEDS+CaixQ4Rf/CseW+VUsxGNEm+emnstVadfyGYZhDullOmTIVR OjBJDMVDCZP+4pBNm807+2B/cOWQRHD+6dwNn+2pQwnfVh+qCJvr8FnlEnuK3fh apcw=;; dara=google.com
ARC-Authentication-Results	i=1; mx.google.com; dkim=pass header.i=@promo.incredunlisted.com header.s=n2048 header.b=soJc20oP; dkim=pass header.i=@env.etransmail.com header.s=fnc header.b=AYzug+VN; spf=pass (google.com: domain of 17872506027708540-200596-1-gmail.com@delivery.promo.incredunlisted.com designate s 103.52.180.244 as permitted sender) smtp.mailfrom=17872506027708540-200596-1-gmail.com@delivery.promo.incredunlisted.com; dmarc=pass (p=QUARANTINE sp=QUARANTINE ds=NONE) header.from=incredunlisted.com
Return-Path	<17872506027708540-200596-1-gmail.com@delivery.promo.incredunlisted.com>
Received-SPF	pass (google.com: domain of 17872506027708540-200596-1-gmail.com@delivery.promo.incredunlisted.com designates 103.52.180.244 as permitted sender) client-ip=103.52.180.244;
Authentication-Results	mx.google.com; dkim=pass header.i=@promo.incredunlisted.com header.s=n2048 header.b=soJc20oP; dkim=pass header.i=@env.etransmail.com header.s=fnc header.b=AYzug+VN; spf=pass (google.com: domain of 17872506027708540-200596-1-gmail.com@delivery.promo.incredunlisted.com designates 103.52.180.244 as permitted sender) smtp.mailfrom=17872506027708540-200596-1-gmail.com@delivery.promo.incredunlisted.com; dmarc=pass (p=QUARANTINE sp=QUARANTINE ds=NONE) header.from=incredunlisted.com
DKIM-Signature	v=1; a=rsa-sha256; c=relaxed/relaxed; d=promo.incredunlisted.com; s=n2048; h=feedback-id:list-unsubscribe-post:list-unsubscribe:content-type:mime-version: subject:message-id:to:from:from:to:subject; bh=Jqh7FzrXoQ4sQeDWZd10xxokrp4oIUJ2YP257Pj8=; b=soJc20oPjYrBmU7KLwg276Y63r33QbXKqN Pm8PehAU9VdC8xzyVbiGMenWuDRR0oGZDC 113BuwoOyikdTJH2L9Rt8d3X4d==
Date	Fri, 21 Aug 2026 19:07:03 +0530
From	InCred Unlisted <marketing@promo.incredunlisted.com>
To	saikiranmutyala18@gmail.com
Message-ID	<1077446648.1550691.1787315557555@ip-10-12-53-116.ap-south-1.compute.internal>
Subject	Orbis Financial Corporation Ltd. Unlisted Shares
MIME-Version	1.0
Content-Type	multipart/alternative; boundary="-----_Part_1550690_2082029169.1787315557555"
List-Unsubscribe	<https://in1.clevertap-prod.com/click?hosting_type=system&one_click=true&sub=0&p=LHg7iYSCxQseHYijcvLJ8XloweGhrdWxmTQJQE07enAImZRcJULExA5Lz8MigeJfXEXEw53eGR%3FYWYrXU8cERYmMYQk13hoaxBHS0Y9BTYUmtSnJ0YSHRR10Qc4Dnha0kSFVoL2AuJRY9JVUSCXR1anAZK%3D%3D&e=2m1q6QMBABIE16YGJieAUAAnRWB7ZwIcwaABLK2M2kUts8JndESFU702Nf20z2Fk0Ic14N00DAXce&encoded=true&wzrk_ex=0>
List-Unsubscribe-Post	List-Unsubscribe=One-Click
X-InjTime	1787315557
Feedback-ID	MJAwnTK2OjwMJYwODixXzE40g==pepiPostE
X-FNCID	200596-17872506027708540-0
X-Traffic-Type	200596-3
X-Mailer	NelcoreCloud Mailer